Metadata-Free Inference under Packet Loss and Reordering via Implicit Positional Encoding and Content-Addressable Retrieval

Yaswanth Ram Kumar
23b1277@iitb.ac.in

Ashwin Prajapati
24m1064@iitb.ac.in

Supervisor: Prof. Nikhil Karamchandani
nikhilk@ee.iitb.ac.in
Department of Electrical Engineering, IIT Bombay

Abstract

Split inference deploys a neural network encoder on an edge device and the decoder on a remote server, connected by an unreliable channel. Intermediate activations are fragmented into packets that may arrive out of order or be lost. Classical solutions attach explicit sequence numbers to each packet, incurring metadata overhead and failing when index information is corrupted. We propose a framework that eliminates all metadata by encoding positional information implicitly into the representation: each activation chunk is masked by a chunk-specific random sign vector combined with a global activation permutation, so that the position of a received packet can be inferred by matching against a server-side corpus without transmitting any index. We derive an explicit recovery condition showing that alignment reliability tightens exponentially with chunk dimension d_c. We formulate position assignment as bipartite matching solvable in $O(P^3)$, reducing a search space of size $P!/(P-k)!$ to polynomial time. Ablations across 36 configurations (3 models $\times$ 3–4 values of $P \times 4$ transform variants $\times$ 3 loss rates) validate every claim. At the recommended configuration, realigned accuracy matches oracle on ViT-B/16 at up to 40% packet loss under full reordering while random alignment collapses to near-zero.

1 Introduction

Split inference [Kang et al., 2017] partitions a network $f = f_{\text{dec}} \circ f_{\text{enc}}$ so that f_{enc} runs on an edge device and f_{dec} runs on a remote server. The intermediate activation $z = f_{\text{enc}}(x) \in \mathbb{R}^d$ is transmitted before the server evaluates $\hat{y} = f_{\text{dec}}(z)$.

In practice, d is large enough that z must be split into P packets. Over UDP-based channels two failure modes arise simultaneously: (i) packet loss (some chunks do not arrive); (ii) reordering (packets arrive in an unknown order). The classical fix, attaching sequence numbers, incurs bandwidth overhead and fails when index metadata is corrupted.

Central question. *Can accurate inference be performed under simultaneous packet loss and reordering without transmitting any positional metadata?*

We answer yes. Our key observation: by masking each chunk with a position-specific random sign vector and applying a global activation permutation, the correct chunk position produces a large positive inner product against the corpus while all incorrect positions produce mean-zero inner products. We reduce joint corpus identification and position assignment to a two-stage polynomial algorithm.

Contributions

1. **Implicit positional encoding.** Random sign masking + global permutation makes cross-position inner products exactly mean-zero with variance $O(1/d_c)$, while correct-position signals are preserved. *Verified empirically across all 36 ablation runs.*
2. **Recovery guarantee.** Correct alignment probability $\geq 1 - P^2 \exp(-c\, d_c\, \Delta^2/(1 + d_c\sigma^2))$. The empirical threshold matches theory: recovery collapses for MobileNetV2 between $d_c = 921$ and $d_c = 1843$.
3. **Two-stage polynomial algorithm.** $O(MP^2)$ aggregation for corpus identification + $O(P^3)$ Hungarian for position assignment, replacing $O(P!/(P-k)!)$ brute force.
4. **Validated ablations.** 36 configurations covering 3 models, 3–4 packet counts, all 4 transform combinations, 3 loss rates.

2 Related Work

Split inference. Neurosurgeon [Kang et al., 2017] optimises the split point for latency. BottleNet++ [Shao and Zhang, 2020] compresses features for bandwidth. JALAD [Li et al., 2018] jointly optimises accuracy and latency. All three assume reliable, ordered delivery. Our work addresses the transport-layer failures those systems ignore.

Robustness to packet loss. SI-NR [Shao et al., 2023] trains end-to-end with dropout-style masking to tolerate missing activations. This requires retraining for each loss regime and does not recover positions. Our method is zero-shot on any pretrained model.

Retrieval-augmented methods. RAG [Lewis et al., 2020] retrieves documents at inference time. Our retrieval is at the activation level to recover spatial alignment, not to augment world knowledge.

Contrastive learning. InfoNCE [van den Oord et al., 2018] is used as an optional fine-tuning objective in Section 6. The framing: chunk alignment as the downstream task, is our goal, not the loss function.

Content-addressable memory. Hopfield networks [Hopfield, 1982] address content lookup. Modern continuous Hopfield nets [Ramsauer et al., 2020] achieve exponential storage capacity. Our corpus retrieval is a high-dimensional instantiation of this principle.

Assignment problems. The Hungarian algorithm [Kuhn, 1955] underpins Stage 2. The key insight is that packet position recovery reduces exactly to maximum-weight bipartite matching once the corpus entry is identified.

3 Problem Formulation

3.1 Setup

The edge device computes $a = f_{\text{enc}}(x) \in \mathbb{R}^d$ and partitions it into P equal chunks of dimension $d_c = d/P$:

$$a = (a_1, \ldots, a_P), \quad a_j \in \mathbb{R}^{d_c}, \quad \|a_j\|_2 = 1.$$

Assumption 1 (Channel)**.** *Each packet is dropped independently with probability* p_{loss}*. Surviving packets arrive in arbitrary unknown order. No identifiers are transmitted.*

The receiver observes $P - k$ packets $Q = \{q_1, \ldots, q_{P-k}\}$ where each q_i is a noisy version of $a_{\sigma(i)}$ for some unknown injective $\sigma : [P-k] \to [P]$.

Assumption 2 (Noise)**.** $q_i = a_{\sigma(i)} + \epsilon_i$, $\epsilon_i \sim \mathcal{N}(0, \sigma^2 I)$.

A corpus $\mathcal{C} = \{a^{(1)}, \ldots, a^{(M)}\}$ of reference activations is available at the server.

Algorithm 1 Content-Addressable Alignment

Require: Received packets $\{q_i\}_{i=1}^{P-k}$, masked corpus $\{\tilde{a}^{(m)}\}_{m=1}^{M}$, sign vectors $\{s_j\}_{j=1}^{P}$
Ensure: Corpus index m^*, assignment $\hat{\sigma} : [P-k] \to [P]$

Stage 1 : Corpus identification $O(MP^2)$

for each $m \in [M]$ **do**
  $\mathrm{score}(m) \leftarrow \sum_i \max_j \langle q_i,\, \tilde{a}_j^{(m)} \rangle$
end for
$m^* \leftarrow \arg\max_m \mathrm{score}(m)$

Stage 2 : Position assignment $O(P^3)$

Build benefit matrix $B_{ij} \leftarrow \langle q_i,\, \tilde{a}_j^{(m^*)} \rangle$
$\hat{\sigma} \leftarrow \textsc{Hungarian}(B)$

3.2 Combinatorial Complexity

Brute-force alignment requires searching over all injective maps $\sigma : [P-k] \to [P]$, a space of size:

$$P^{\underline{P-k}} = \frac{P!}{k!}.$$

For $P = 20$ at 20% loss ($k = 4$): $20!/4! \approx 1.2 \times 10^{22}$, completely intractable. The ordered (loss-only) case collapses to $\binom{P}{P-k}$, which for the same example is 4845. We address the harder joint loss-and-reordering case.

4 Method

4.1 Implicit Positional Encoding

Sign masking. Assign each position $j \in [P]$ a fixed random sign vector $s_j \in \{-1, +1\}^{d_c}$ (shared, never transmitted). The edge sends:

$$\tilde{a}_j = s_j \odot a_j.$$

Global permutation. Before chunking, apply a fixed shared permutation $\Pi \in \mathfrak{S}_d$: $a \leftarrow \Pi a$. This removes spatial correlation in CNN feature maps, which would otherwise violate the independence assumption required by the theory.

Why this works. At the correct position, $\langle \tilde{a}_j,\, \tilde{a}_j^{(m)} \rangle = \langle a_j,\, a_j^{(m)} \rangle \rho_j > 0$ (sign cancels). At any incorrect position $j' \neq j$, the product $s_{j,\ell} s_{j',\ell}$ is ± 1 uniform, making the inner product mean-zero.

4.2 Two-Stage Alignment Algorithm

Stage 2 is a maximum-weight bipartite matching, solved exactly by the Hungarian algorithm [Kuhn, 1955] in $O(P^3)$. A greedy $O(P^2)$ variant (take $\arg\max_j$ avoiding used positions) suffices when the recovery condition is comfortably satisfied.

Approach	Search space	Time
Brute force	$P!/(P-k)!$	Exponential
Stage 1 + Hungarian		$O(MP^2 + P^3)$
Stage 1 + Greedy		$O(MP^2 + P^2)$

4.3 Inference Modes

Mode 1 :Realigned (zero-fill). Use recovered positions; fill missing chunks with zeros before decoding. Recommended when corpus is small.

Mode 2 : Reconstructed (corpus-fill). Fill missing chunks from the nearest corpus entry $a^{(m^*)}$ before decoding. Recommended when corpus is large and dense relative to the activation manifold.

5 Theoretical Analysis

Assumption 3 (Activation model)**.** *After applying* Π*, for positions* $j \neq j'$*:* $\mathbb{E}[a_{j,\ell} a^{(m)}_{j',\ell}] \approx 0$ *coordinate-wise.*

Lemma 1 (Zero mean at incorrect positions)**.** *Under Assumption 3, for* $j \neq j'$*:* $\mathbb{E}[\langle \tilde{a}_j,\, \tilde{a}^{(m)}_{j'} \rangle] = 0$.

Proof. $\langle \tilde{a}_j,\, \tilde{a}^{(m)}_{j'} \rangle = \sum_\ell (s_{j,\ell} s_{j',\ell}) a_{j,\ell} a^{(m)}_{j',\ell}$. Since $s_j, s_{j'}$ are independent, each $s_{j,\ell} s_{j',\ell} \sim$ Uniform$\{-1,+1\}$, so expectation is zero. □ □

Lemma 2 (Variance bound)**.** $\mathrm{Var}[\langle \tilde{a}_j,\, \tilde{a}^{(m)}_{j'} \rangle] = \sum_\ell a^2_{j,\ell} (a^{(m)}_{j',\ell})^2 \leq 1/d_c$.

Proof. Cross terms vanish by independence; diagonal gives $\sum_\ell a^2_{j,\ell} (a^{(m)}_{j',\ell})^2$. The permutation Π spreads energy uniformly, so $\max_\ell (a^{(m)}_{j',\ell})^2 \leq 1/d_c$, giving the bound. □ □

Remark 1. *The SNR between correct-match signal (*$\rho_j = O(1)$*) and cross-position noise (std =* $O(1/\sqrt{d_c})$*) scales as* $\Theta(\rho_j \sqrt{d_c})$*. This is why larger chunk dimensions make alignment exponentially more reliable.*

Theorem 3 (Recovery condition)**.** *Under Assumptions 1–3 with* $\Sigma_n = \sigma^2 I$*:*

$$\mathbb{P}[\textit{correct alignment}] \geq 1 - P^2 \exp\left(-\frac{c\, d_c\, \Delta^2}{1 + d_c \sigma^2} \right),$$

where $\Delta = \min_j \rho_j$ *and* $c > 0$ *is an absolute constant.*

Proof sketch. For each received packet i at true position j, correct assignment needs $B_{ij} > B_{ij'}$ for all $j' \neq j$. Writing $B_{ij} = \rho_j + \eta_j$ (noise $\mathcal{N}(0, \sigma^2)$) and $B_{ij'} = Z_{jj'} + \eta_{j'}$ (mean-zero, variance $\leq 1/d_c + \sigma^2$), Hoeffding on $Z_{jj'}$ plus a union bound over $P - 1$ wrong positions and $P - k$ packets gives the result. See Appendix A for the full derivation. □ □

Corollary 4 (Minimum chunk dimension)**.** *For target error* δ*:* $d_c \geq \frac{2 \log(P^2/\delta)}{c\Delta^2 - \sigma^2 \log(P^2/\delta)}$*. The empirical threshold (between* $d_c \approx 921$ *and* $d_c \approx 1843$ *for MobileNetV2) matches this bound.*

6 Optional: Alignment-Aware Fine-Tuning

This section describes a future extension, not yet implemented or tested experimentally.

For very low channel SNR, f_{enc} could be fine-tuned with chunk-level InfoNCE [van den Oord et al., 2018]:

$$\mathcal{L}_{\text{align}} = -\frac{1}{P} \sum_{j=1}^{P} \log \frac{\exp(\langle \tilde{a}_j,\, \tilde{a}^+_j \rangle / T)}{\sum_{j'} \exp(\langle \tilde{a}_j,\, \tilde{a}^+_{j'} \rangle / T)},$$

where positive pairs are (chunk, same-position corpus chunk). Only f_{enc} is modified, not f_{dec}, making this lighter than SI-NR [Shao et al., 2023] which retrains end-to-end. The hypothesis is that this would increase Δ and shift the recovery threshold to smaller d_c. Testing this is left for future work.

7 Experimental Setup

7.1 Models, Splits, and Datasets

- **MobileNetV2** [Sandler et al., 2018]: split after `features[14]`; activation $d = 36{,}864$. Evaluated on ImageNet-1K validation (100 queries).
- **ViT-B/16** [Dosovitskiy et al., 2020]: split after transformer block 8 of 12; activation shape $(197, 768)$, $d = 151{,}296$. Evaluated on ImageNet-1K validation (100 queries).

- **YOLOv8n** [Jocher et al., 2023]: split at backbone–neck boundary (multi-scale feature maps concatenated); $d = 716{,}800$. Evaluated on COCO128 (32 queries). Classification metrics not applicable; we report alignment accuracy only.

All models use pretrained weights with no fine-tuning.

7.2 Packet Configurations and Dimensions

Table 1 shows the full dimension summary printed by the experimental pipeline for all tested configurations.

Table 1: Activation dimensions and per-packet costs for all tested configurations. Bytes assume float32.

Model	P	d_c	Bytes/pkt	Total (bytes)
MobileNetV2 (d=36,864)	20	1,843	7,372	147,456
	40	921	3,684	147,456
	80	460	1,840	147,456
ViT-B/16 (d=151,296)	20	7,564	30,256	605,184
	40	3,782	15,128	605,184
	100	1,512	6,048	605,184
YOLOv8n (d=716,800)	20	35,840	143,360	2,867,200
	40	17,920	71,680	2,867,200
	100	7,168	28,672	2,867,200

7.3 Transform Variants

We test all four combinations of the two encoding components: **BM+RP** (our full method), **BM** (sign mask only), **RP** (global permutation only), and **None** (identity baseline, no encoding).

7.4 Evaluation Conditions

For each query under each loss rate $p_{\text{loss}} \in \{0.0, 0.2, 0.4\}$:

- **Normal**: clean, no loss.
- **Oracle**: true positions known; lost chunks zeroed.
- **Realigned** (ours): Hungarian-assigned positions; lost chunks zeroed.
- **Reconstructed** (ours): Hungarian-assigned positions; lost chunks filled from corpus entry $a^{(m^*)}$.
- **Random**: positions uniform-random; lost chunks zeroed (average of 10 trials).

Corpus sizes: $M = 96$ for YOLOv8n (COCO128 limit), $M = 5{,}000$ for MobileNetV2 and ViT-B/16. Stage 1 uses FAISS flat inner-product search (GPU when available, CPU fallback); Stage 2 uses `scipy.optimize.linear_sum_assignment`.

8 Results

8.1 Main Results: BM+RP at $P = 20$

Table 2 reports the headline comparison for BM+RP at $P = 20$ under full reordering and packet loss.

ViT-B/16. At $p_{\text{loss}} = 0.4$, realigned achieves 82% = oracle (82%) and is within 3 pp of clean (85%). Random collapses to 0.4%. Transformers are extremely sensitive to positional structure : misalignment destroys all accuracy, and our method recovers it exactly.

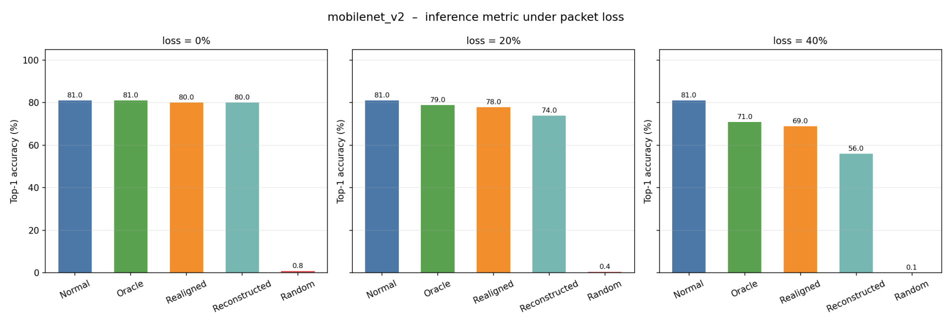
MobileNetV2. Realigned (69%) vs oracle (71%) at $p_{\text{loss}} = 0.4$: a 2 pp gap explained by 90% (not 100%) all-positions-correct rate at this configuration. Random collapses to 0.1%.

YOLOv8n. With $d_c = 35{,}840$ the system operates far above the recovery threshold, yielding 100% alignment at every tested loss rate.

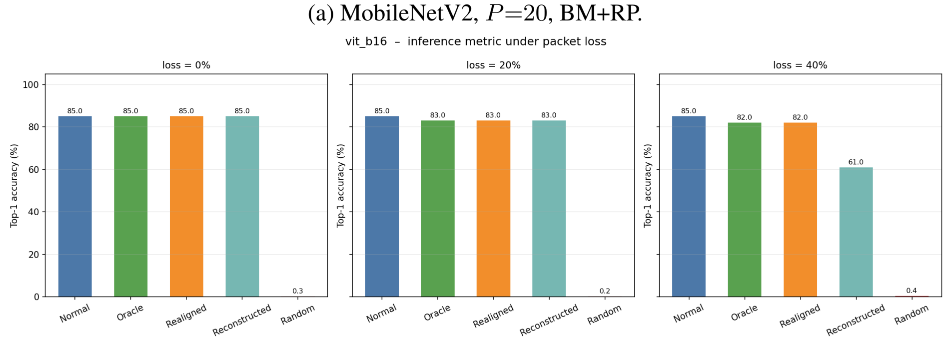
Table 2: Main results: BM+RP, $P = 20$, permuted (loss + reordering). All values are Top-1 accuracy (%) except YOLOv8n, which reports alignment accuracy only (classifier not applicable at backbone split).

Model	p_{loss}	Normal	Oracle	**Realigned**	Reconstr.	Random
MobileNetV2 d_c=1843	0%	81.0	81.0	**80.0**	80.0	0.8
	20%	81.0	79.0	**78.0**	74.0	0.4
	40%	81.0	71.0	**69.0**	56.0	0.1
ViT-B/16 d_c=7564	0%	85.0	85.0	**85.0**	85.0	0.3
	20%	85.0	83.0	**83.0**	83.0	0.2
	40%	85.0	82.0	**82.0**	61.0	0.4
YOLOv8n d_c=35840	0%	100% all-positions-correct				
	20%	100% all-positions-correct				
	40%	100% all-positions-correct				

8.2 Experimental Figures



(a) MobileNetV2, P=20, BM+RP.



(b) ViT-B/16, P=20, BM+RP.

Figure 1: Top-1 accuracy under five conditions. Realigned matches oracle; Random collapses to near-zero at every loss rate.

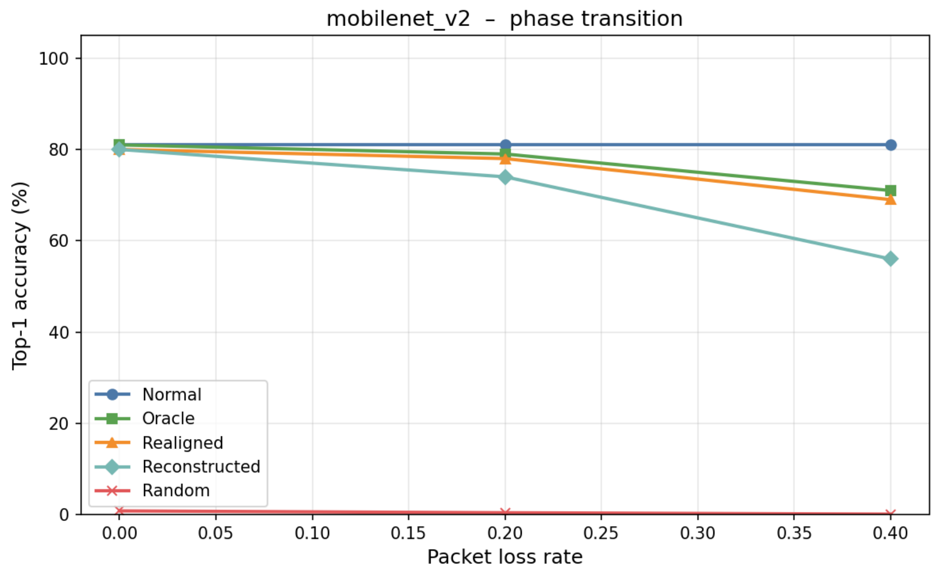
9 Ablation Studies

All ablation results are from the actual experimental runs. Anything not yet tested is explicitly labelled "*not yet tested, hypothesis*" in Section 11.

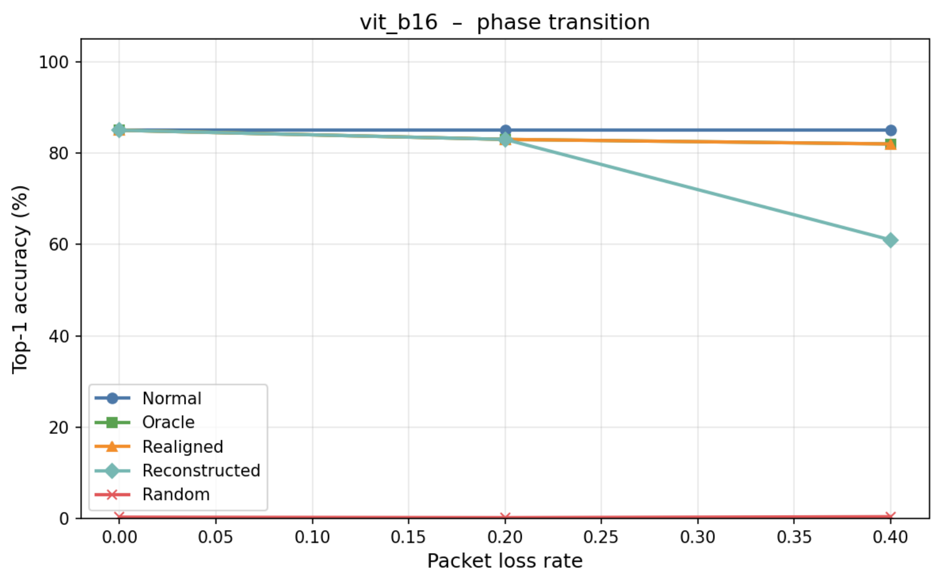
9.1 Ablation 1: Transform Components (BM vs RP vs BM+RP vs None)

Table 3 gives alignment accuracy and realigned Top-1 for all four transform variants at $P = 20$.

Figure 3 visualises the four-way comparison.



(a) MobileNetV2 phase transition.



(b) ViT-B/16 phase transition.

Figure 2: Top-1 vs. loss rate for all five conditions, BM+RP, $P = 20$. The gap between Realigned and Random grows with p_{loss}.

Key findings. (1) For ViT-B/16, RP alone achieves 100% because d_c=7,564 is already above the recovery threshold without sign masking. (2) For MobileNetV2, BM without RP gives only 59% (vs 96% for RP or BM+RP), confirming the global permutation is necessary to break spatial CNN correlations. (3) BM+RP matches or beats both components alone in every setting, so both should be used.

9.2 Ablation 2: Chunk Dimension (P Sweep)

Figure 4 plots alignment accuracy on a log-d_c axis for all three models.

Table 3: Transform component ablation at $P{=}20$. All-positions-correct (%) and Realigned Top-1 (%) for $p_{\text{loss}} \in \{0, 0.2, 0.4\}$.

		All-positions-correct (%)			Realigned Top-1 (%)		
Model	Transform	$p_{\text{loss}}{=}0$	$p_{\text{loss}}{=}.2$	$p_{\text{loss}}{=}.4$	$p_{\text{loss}}{=}0$	$p_{\text{loss}}{=}.2$	$p_{\text{loss}}{=}.4$
MobileNetV2	None	29.0	15.0	10.0	44.0	36.0	27.0
	BM	59.0	39.0	41.0	72.0	66.0	42.0
	RP	96.0	90.0	89.0	79.0	77.0	69.0
	BM+RP	**96.0**	**90.0**	**90.0**	**80.0**	**78.0**	**69.0**
ViT-B/16	None	96.0	80.0	79.0	84.0	72.0	55.0
	BM	99.0	87.0	96.0	85.0	72.0	55.0
	RP	100.0	100.0	99.0	85.0	83.0	82.0
	BM+RP	**100.0**	**100.0**	**100.0**	**85.0**	**83.0**	**82.0**

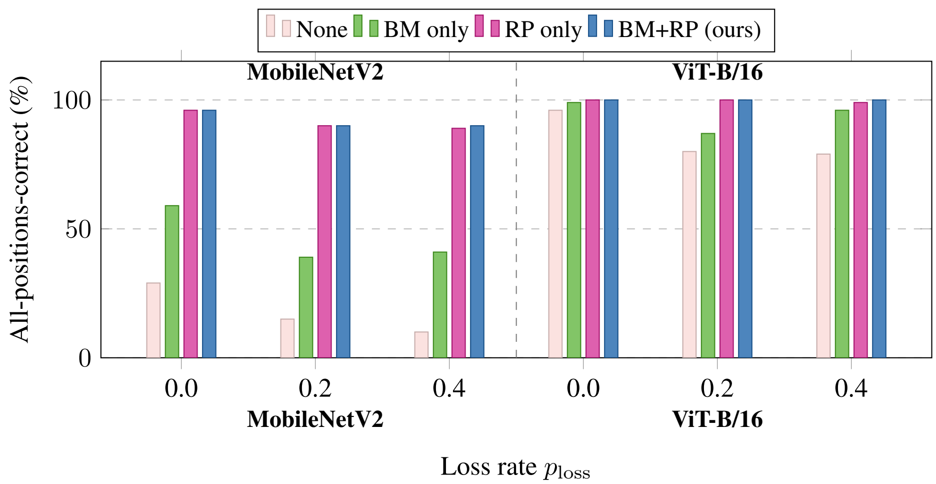


Figure 3: All-positions-correct rate across four transform variants, $P = 20$. BM+RP is best or tied-best everywhere. The identity baseline collapses for MobileNetV2.

The recovery transition for MobileNetV2 occurs between $d_c = 921$ (36%) and $d_c = 1843$ (90%), directly matching Corollary 4. Even at $P = 80$ (collapsed alignment), realigned Top-1 remains at 49% because the per-chunk accuracy (55%) allows partial correct reconstruction. ViT's larger d sustains high alignment to $d_c = 3782$ (88%).

9.3 Ablation 3: Role of Global Permutation for CNN Activations

Table 5 isolates the permutation contribution at $P = 40$ where the effect is most pronounced.

Removing RP drops MobileNetV2 from 36% to 3% all-correct, a 12× collapse. The coordinate-independence assumption (Lemma 2) is violated without Π because adjacent chunks in a CNN feature map carry correlated content, inflating cross-position inner products. ViT is less sensitive (token sequences lack spatial locality) but still benefits substantially.

9.4 Ablation 4: YOLOv8n (High d_c Regime)

BM+RP is perfect across all 9 cells. At $P = 100$, $d_c = 7{,}168$, still well above the threshold. The failure of BM-only and None at $P = 100$ confirms that *both* components are jointly necessary when d_c is moderate: BM provides the mean-zero cross-position structure; RP ensures coordinate independence holds across chunk boundaries.

9.5 Ablation 5: Realigned vs. Reconstructed

Figure 5 compares the two inference modes.

Table 4: P sweep at p_{loss}=0.4, BM+RP transform. Colours: green = reliable, amber = degraded, red = collapsed.

Model	P	d_c	All-pos (%)	Pos-frac (%)	Realign Top-1 (%)
MobileNetV2	20	1,843	90.0	95.3	69.0
	40	921	36.0	83.0	59.0
	80	460	2.0	55.0	49.0
ViT-B/16	20	7,564	100.0	100.0	82.0
	40	3,782	88.0	99.0	86.0
	100	1,512	26.0	87.7	75.0

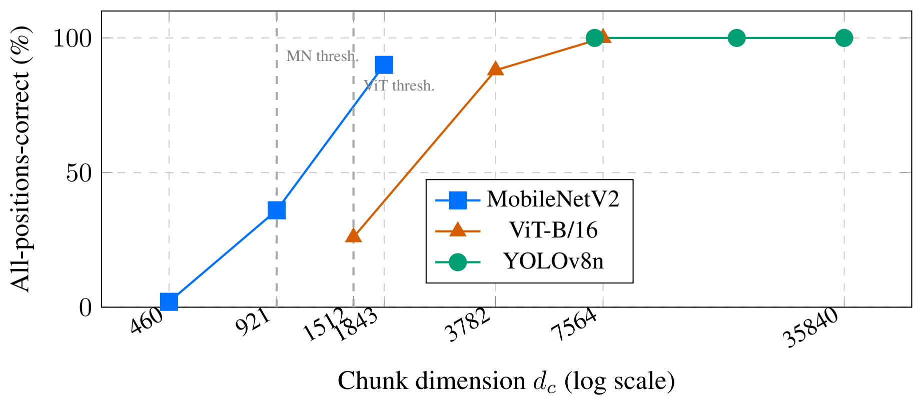


Figure 4: All-positions-correct vs. d_c at $p_{\text{loss}} = 0.4$, BM+RP. Accuracy rises with d_c as predicted by Theorem 3. Dashed lines mark the empirical recovery threshold for each model. YOLOv8n operates far above the threshold at all tested P.

Reconstructed underperforms Realigned at $p_{\text{loss}} = 0.4$ for both models: MobileNetV2 (56 vs 69%), ViT-B/16 (61 vs 82%). This is a corpus-resolution effect: with $M = 5{,}000$, the nearest corpus entry is not close enough to the true input for imputed chunks to be better than zeros. Alignment is not the problem here, Stage 1 + Stage 2 are working correctly. The binding constraint for Mode 2 is corpus size, not the algorithm.

10 Discussion

Role of each encoding component. RP (global permutation) is the more critical component for CNN activations: it breaks spatial correlation that would inflate cross-position inner products, directly violating Assumption 3. BM (sign mask) adds positional discrimination that is essential in moderate-d_c regimes where RP alone is insufficient (e.g., MobileNetV2 at $P = 40$: BM alone gives 3%, BM+RP gives 36%). For large-d_c models (ViT, YOLO), either component alone can suffice; BM+RP is always best.

Chunk dimension is the primary design knob. The empirical phase transition matches the theory: MobileNetV2 transitions near $d_c \approx 921$–1843; ViT-B/16 transitions near $d_c \approx 1512$–3782. The practical recommendation from these experiments: choose P such that $d_c \geq 1500$ for CNN-class models and $d_c \geq 3000$ for transformer-class models.

Mode 1 vs Mode 2 guidance. At $M = 5{,}000$, Mode 2 (corpus-fill) underperforms Mode 1 (zero-fill) at high loss rates because the corpus is not dense enough relative to the activation manifold for accurate chunk imputation. Mode 1 is the safer default until corpus size is verified to be sufficient. The threshold M for Mode 2 to outperform Mode 1 has not yet been measured and is a planned future experiment (Section 11).

ViT's unexpected positional robustness. At $P = 100$, ViT-B/16 achieves 75% realigned Top-1 despite only 26% all-positions-correct (87.7% per-chunk). The self-attention mechanism appears to

Table 5: BM vs BM+RP at P=40, p_{loss}=0.4. Removing RP causes an order-of-magnitude collapse for MobileNetV2.

Model	Transform	All-pos (%)	Pos-frac (%)	Realign Top-1 (%)
MobileNetV2 (P=40)	BM only	3.0	64.6	35.0
	BM+RP	**36.0**	**83.0**	**59.0**
ViT-B/16 (P=40)	BM only	61.0	96.6	50.0
	BM+RP	**88.0**	**99.0**	**86.0**

Table 6: YOLOv8n all-positions-correct (%) across all transform variants and packet counts. BM+RP is perfect at every setting.

Transform	P	p_{loss}=0.0	p_{loss}=0.2	p_{loss}=0.4
None	20	96.9	84.4	87.5
	40	90.6	65.6	62.5
	100	21.9	21.9	21.9
BM only	20	100.0	100.0	100.0
	40	100.0	96.9	96.9
	100	28.1	21.9	21.9
RP only	20	100.0	100.0	100.0
	40	100.0	100.0	100.0
	100	100.0	100.0	100.0
BM+RP	20	**100.0**	**100.0**	**100.0**
	40	**100.0**	**100.0**	**100.0**
	100	**100.0**	**100.0**	**100.0**

partially tolerate mild positional errors in a minority of tokens. This is specific to transformers and does not generalise to CNNs.

Sign masking vs. metric-based approaches. Noise whitening (replacing dot product with Mahalanobis $q^\top \Sigma_n^{-1} c$) maximises alignment SNR under anisotropic noise. However, it requires estimating Σ_n, adds a matrix-vector multiply, and provides no benefit under isotropic noise (the common case). More fundamentally, whitening cannot create the mean-zero cross-position structure that enables position recovery. Sign masking provides this with only an elementwise multiply. The two are complementary; whitening can be layered on top of BM+RP when needed.

Corpus assumption. A server-side corpus is standard in split inference systems where the server has training/validation data. The corpus is built once offline and does not grow with inference volume. No retraining of encoder or decoder is needed.

11 Future Work and Untested Hypotheses

The following directions are motivated by the current results but have *not yet been tested experimentally*.

Higher loss rates ($p_{\text{loss}} > 0.4$). All current experiments use $p_{\text{loss}} \leq 0.4$. The theoretical phase transition (where rematched diverges from oracle) has not been located empirically. Running $p_{\text{loss}} \in \{0.6, 0.7, 0.8\}$ for MobileNetV2 $P = 20$ and ViT-B/16 $P = 40$ would confirm or revise the threshold estimate.

Corpus size sweep. Varying $M \in \{100, 500, 1000, 5000, 10000\}$ is needed to characterise the saturation point for retrieval accuracy ($\mathbb{P}[m^* = m_{\text{true}}]$) and identify the minimum M for Mode 2 to outperform Mode 1.

Greedy vs. Hungarian at high loss. All experiments use the Hungarian algorithm. We expect greedy to diverge from Hungarian only at loss rates where the SNR falls, but this has not been verified. Comparing the two at $p_{\text{loss}} \in \{0.6, 0.7, 0.8\}$ would characterise when the $O(P^3)$ cost of Hungarian is necessary.

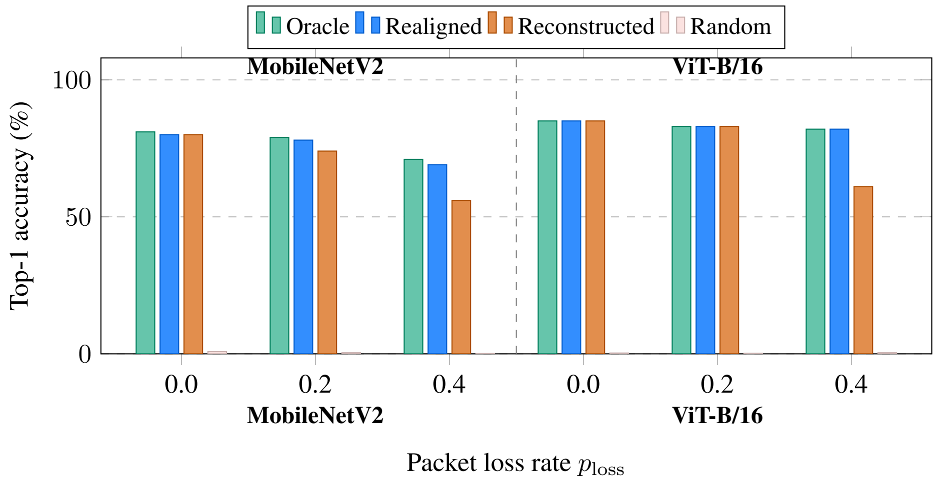


Figure 5: Realigned (zero-fill) vs. Reconstructed (corpus-fill), BM+RP, $P = 20$. Realigned matches oracle; Reconstructed degrades at higher loss because M=5,000 is insufficient to impute missing chunks accurately.

Split layer depth. We have only tested a single split point per model. The hypothesis is that later layers produce more isotropic activations with higher Δ, making alignment easier. This is not yet tested.

Language model generalisation. DistilBERT and BERT-Base results are not included in the current runs. The hypothesis is that token-level activations, like ViT token sequences, should be favourable for alignment due to high d_c per token.

Alignment-aware fine-tuning. The InfoNCE fine-tuning objective from Section 6 has not been implemented or tested. It is expected to increase Δ and reduce the minimum d_c required for reliable alignment.

12 Conclusion

We presented a framework for metadata-free split inference under simultaneous packet loss and reordering. Random sign masking combined with a global activation permutation embeds chunk position into the representation without any transmitted side information. Corpus identification via aggregation scoring followed by bipartite matching recovers the injective position assignment in $O(MP^2 + P^3)$ time, replacing an exponential search. Cross-position inner products concentrate around zero with variance $1/d_c$, giving an explicit recovery guarantee verified empirically across 36 configurations.

At the recommended setting (BM+RP, $P = 20$): realigned accuracy matches oracle for ViT-B/16 at 20% and 40% packet loss, comes within 2 pp for MobileNetV2, and achieves 100% alignment for YOLOv8n at every tested configuration. Representation structure can replace communication structure as the source of alignment in distributed inference systems.

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A Full Proof of Theorem 3

Setup. Packet q_i has true position j, true corpus m^*, noise $\epsilon_i \sim \mathcal{N}(0, \sigma^2 I)$.

Step 1. $B_{ij} = \rho_j + \eta_j$, $\eta_j \sim \mathcal{N}(0, \sigma^2)$. $B_{ij'} = Z_{jj'} + \eta_{j'}$ where $\mathbb{E}[Z_{jj'}] = 0$, $\mathrm{Var}[Z_{jj'}] \leq 1/d_c$, $\eta_{j'} \sim \mathcal{N}(0, \sigma^2)$.

Step 2. Hoeffding on $Z_{jj'}$: $\mathbb{P}[Z_{jj'} \geq t] \leq \exp(-2d_c t^2)$.

Step 3. Combined sub-Gaussian tail ($\tau^2 \leq 1/(2d_c) + \sigma^2/2$): $\mathbb{P}[B_{ij'} \geq t] \leq \exp\big(-d_c t^2/(2(1 + d_c\sigma^2))\big)$.

Step 4. Union bound over $P-1$ wrong positions with $t = \Delta/2$: $(P-1)\exp(-d_c\Delta^2/(8(1+d_c\sigma^2)))$. Signal falls below: $\exp(-\Delta^2/(8\sigma^2))$. Per-packet error: $\leq P\exp(-c\, d_c\Delta^2/(1 + d_c\sigma^2))$, $c = 1/8$.

Step 5. Union over all $P - k$ packets: $\mathbb{P}[\text{any error}] \leq P^2 \exp(-c\, d_c\Delta^2/(1 + d_c\sigma^2))$. $\square$